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Completed

Exp:1 - Study of Various Network Commands used in Linux and Windows

Aim :- To study the various types of network Commands used in linux & windows.

1. **Command** :- arp -a

Description :- ARP is short form of addres resolution protocol, it will show the IP address of your computer with the IP & MAC address of your router.

output :-

Address HW type HW address Elaskmask

gateway.3 ether 52:55:09:DD:02:02 C

2. **Command** :- hostname

Description :- This is the simplest of TCP/IP commands. It simply displays the name of your computer.

output :- fedora

3. **Command** :- ipconfig /all

Description :- This command displays detailed configuration information about your tcp/ip connection including router, gateway, DNS, DHCP.

Output :-

Hostname ..... : Desktop-DPRAGHK

Node Type ..... : Mixed

4. Command :- nbtstat -a

Description :- This command helps solve problem with NetBios name resolution.

Output :-

NBTSTAT [[-A Remote Name] [-A IP ADDRESS] [-c] [-R] [-r] [-RR] [-S] [-s] [interval]]

5. Command :- netstat

Description :- (Network statistics) - displays variety of statistics about computer activity.

Output :-

Active internet connections (w/o servers)

Proto Recv Send a local address

6. Command :- nslookup

Description :- (name servers lookup) is a tool used to perform DNS lookups in Linux

output :-

Server : 127.0.0.53

Address : 127.0.0.53 #53

Name : www.googh.com

7. Command :- pathping

Description :- It is a combination of ping & tracert commands. Pathing traces the route of destination then launches a 25 second test of each routes.

output :-

Usage :- pathping [-g host list] [-h maximum-hops]
[-i address] [-n]
[-p period] [-q num]

8) Command : ping

Description : (Packet Internet Groupes) command is the best way to test connectivity

eg : ping hostname

output :-

ping localhost

PING localhost (::1) 56 data bytes

64 bytes from local host (::1) : cmp-seq =1 ttl = 64

9) Command :- Route

Description :- used to show / manipulate the IP routing tables

output :- Kernel IP routing tables

Destination Gateway Genmask flask metric

default gateway 0.0.0.0 UG 100

10. Command :- IP

Description :- The IP command can show address information, manipulate routing, plus display network.

Eg : ip address show

output :-

1 : lo : <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc

no queue state unknown group default qlen 1000.

11. Command :- enp0s3

Description :- ifaddress add 192.168.1.254/24

12. Command :- Ifconfig

Description :- Displays and configures the network interface settings of a Linux system

output :-

inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255

13. Command :- mtr

Description :- Combines Ping and traceroute to analyze the network path and identify packet loss or latency

output :-

Hostname	Loss	Snt	Last	Avg	Best	Worst
localhost	0.0%	41	0	0	0	0

a)

mtr google.com

output :-

lekk02 0.0% 30 6 27 6 132

b) mtr -g google.com

Pnmaa 1.8 59 6 22 5

c) mtr -b google.com

Pnmaa 0.0 9 11 39 8

14. ~~tcpdump~~

Command :- tcpdump

description :- Captures and analyzes network packets on a selected network interface.

output :-

Snapshot length 262144 Bytes

15. Command :- nmcli

Description :- Command-line tool used to configure and manage Network Manager network Connections

output :- " Intel 82540EM "

ethernet (e1000)

ip4 default, ip6 default

Student observation :-

1) which Command is used to find the reachability of a host machine from your device ?

A) PING

2) which command gives the details of hops taken by a packet to reach its destination ?

A) mtr

3) which command displays the IP configuration of your machine ?

A) ipconfig

4) which command displays the TCP port status in your machine ?

A) tcpdump

5) write the command to modify the IP configuration in a Linux machine.

A) nmcli connection modify "Wired connection 1" ipv4.method auto.

Exp:2 - Study of different types of Network Cables

Aim:- To study the different types of network cables.

a) understand different types of network Cables

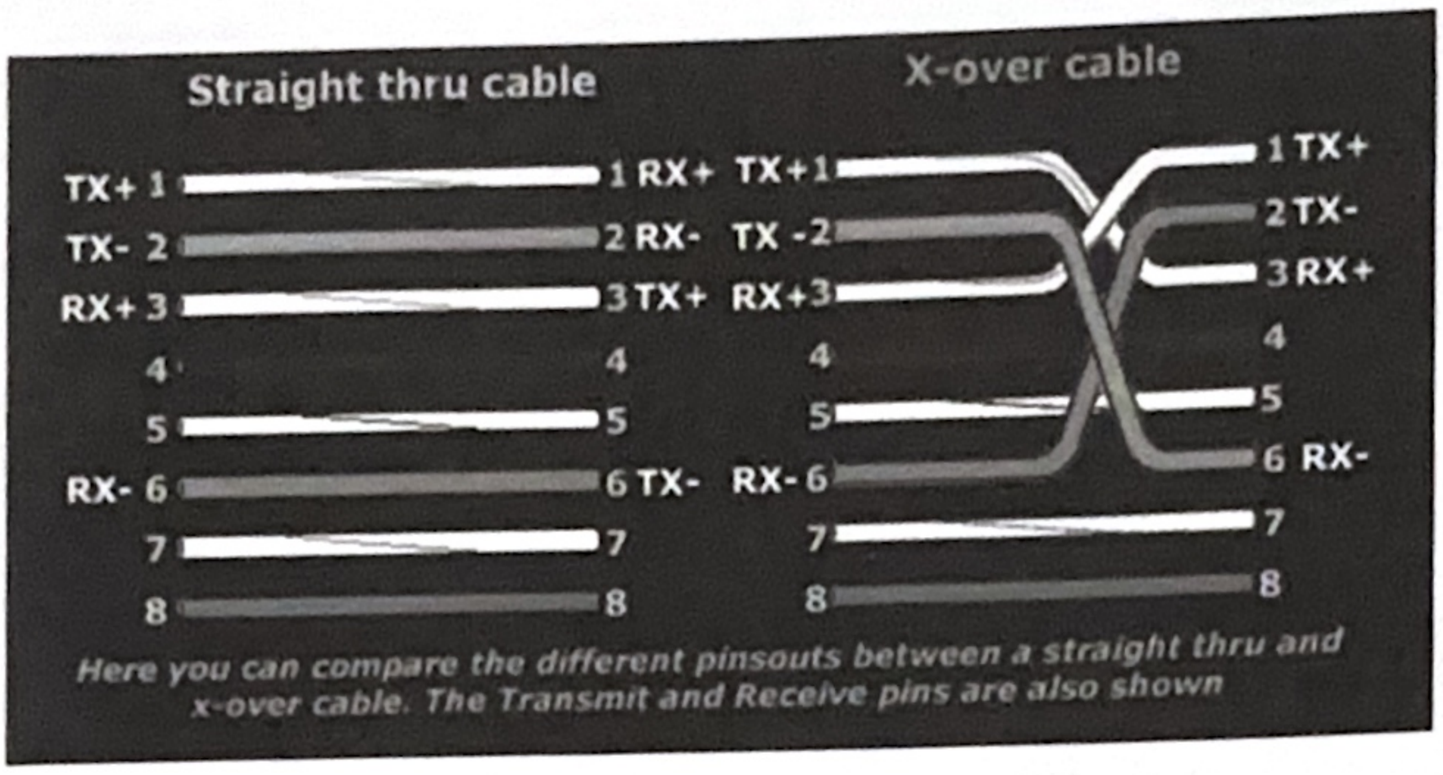
1) unshielded Twisted pair (UTP) Cables.

2) Shielded Twisted pair (STP) Cables.

3) Coaxial cables

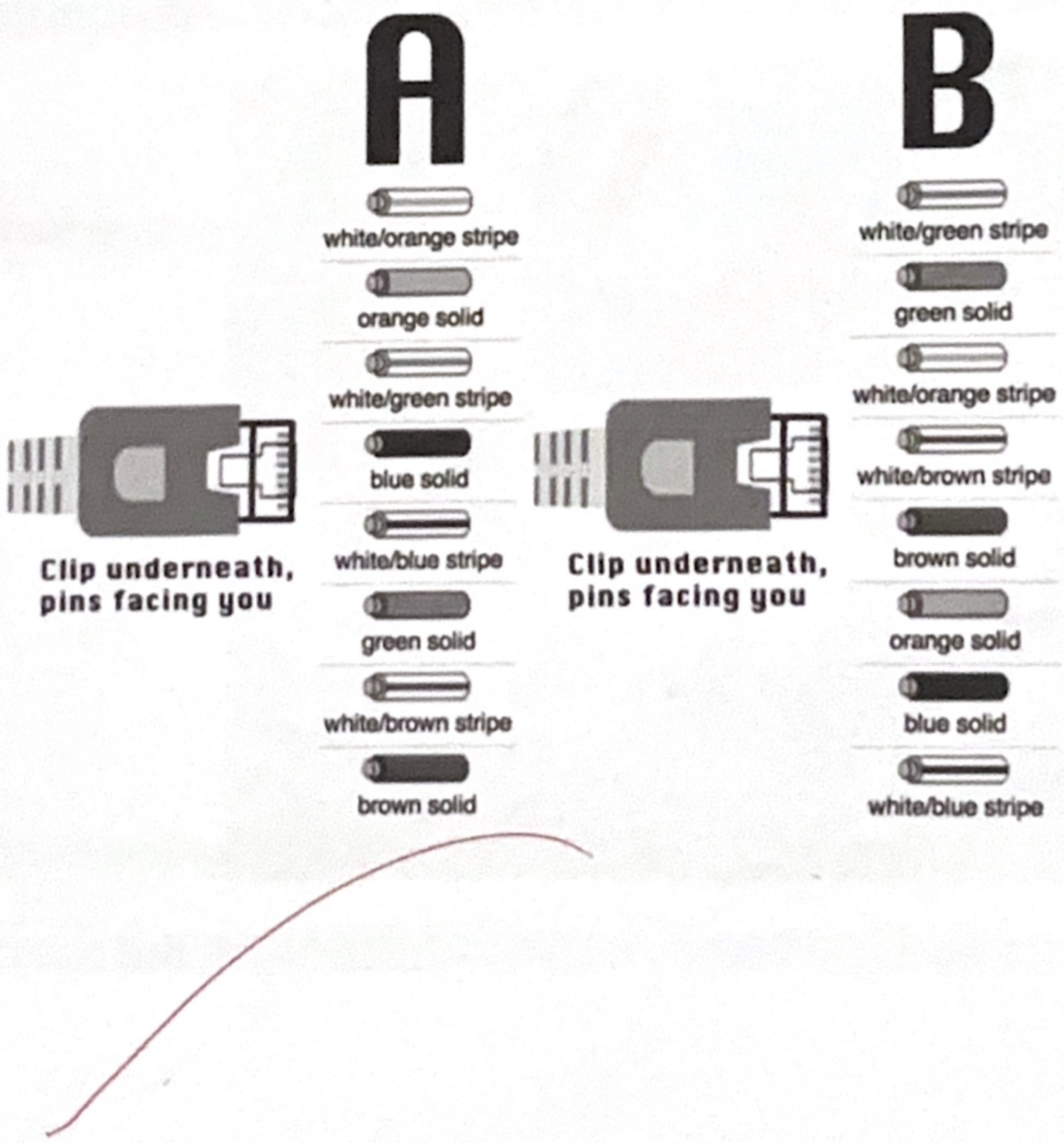
4) Fiber optic Cable

EXPERIMENT - 2



Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

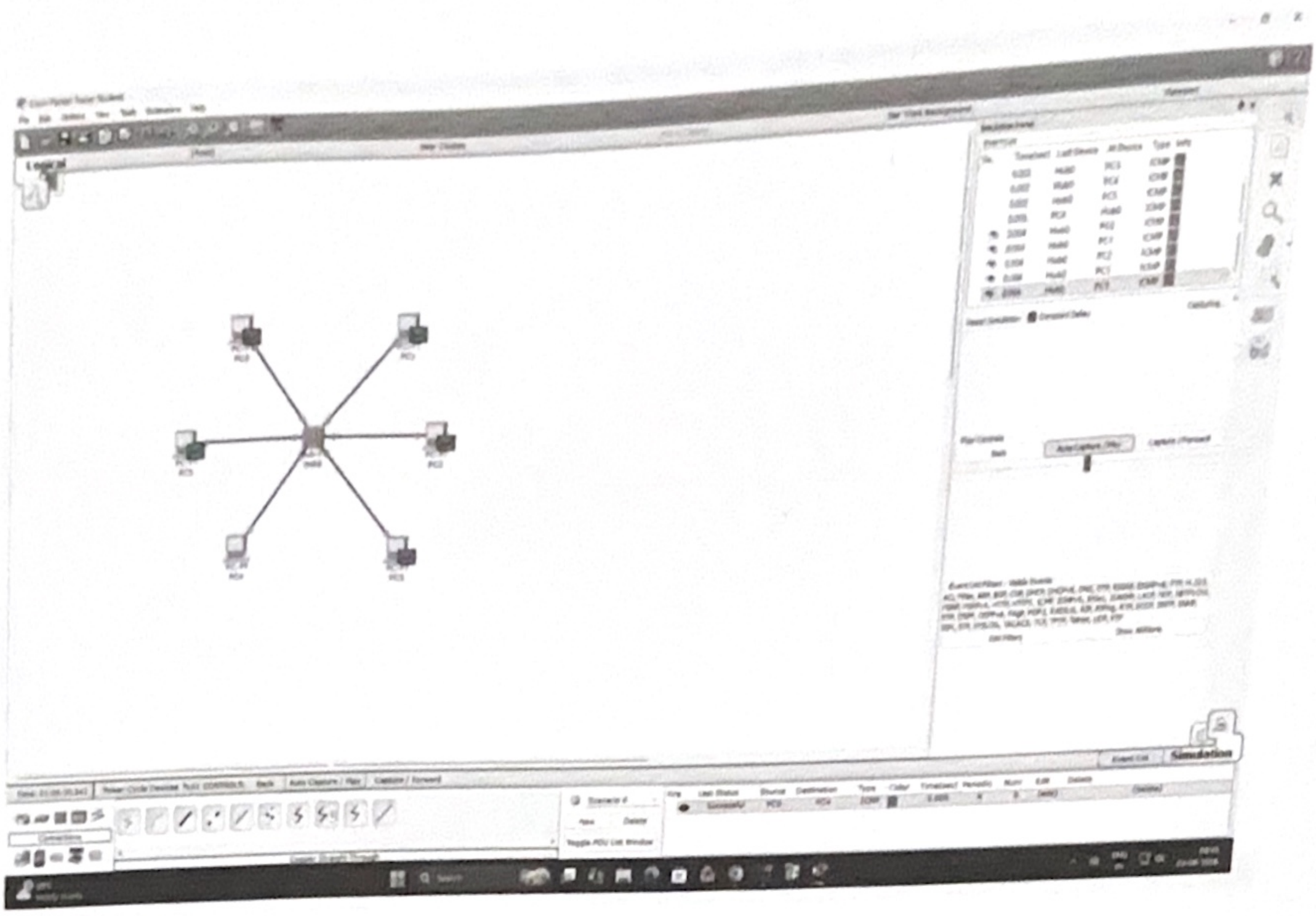
Straight through network cable: both sides should be A
Crossover cable: One side A, one side B

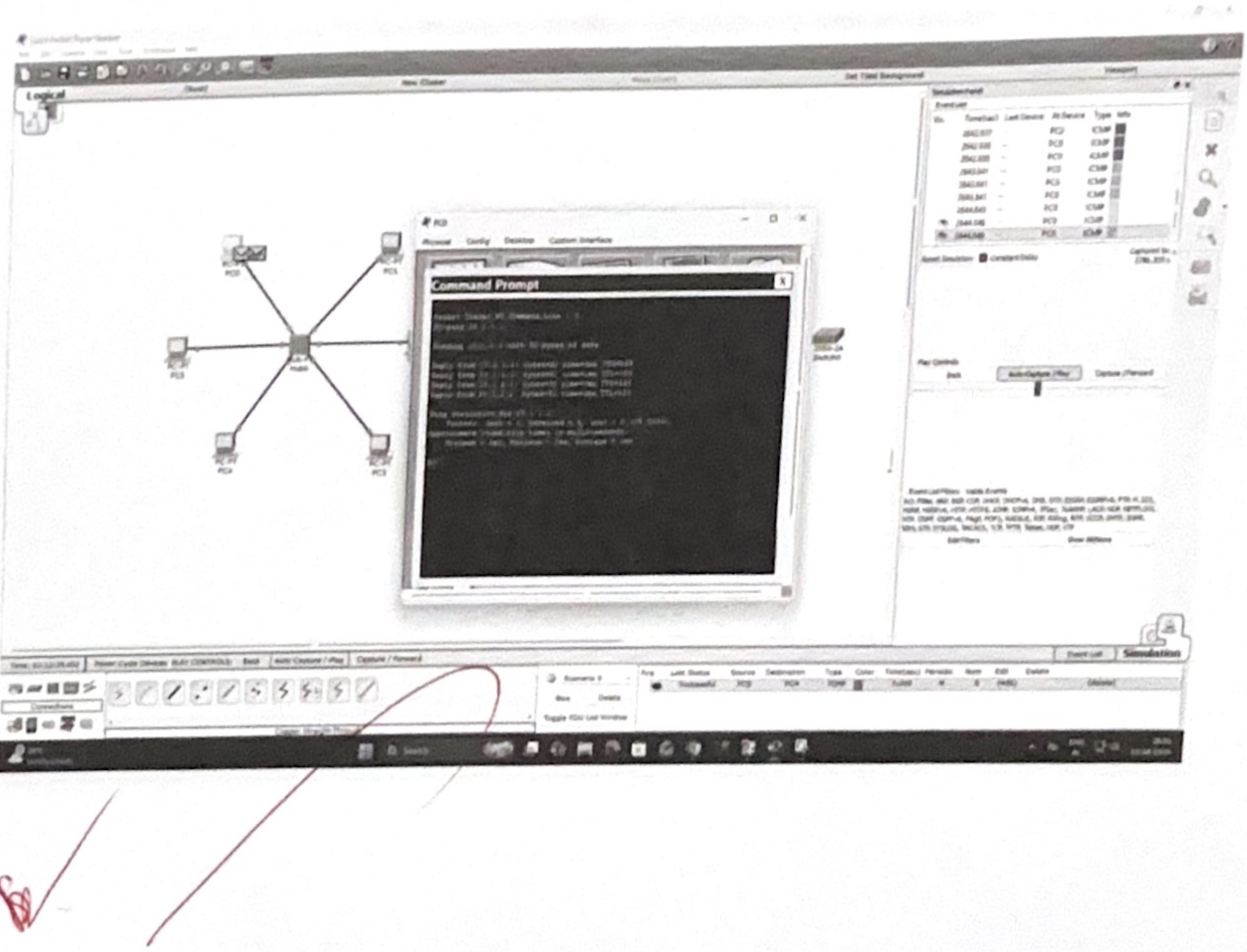



A
B


A
orange solid
blue solid
underneath,

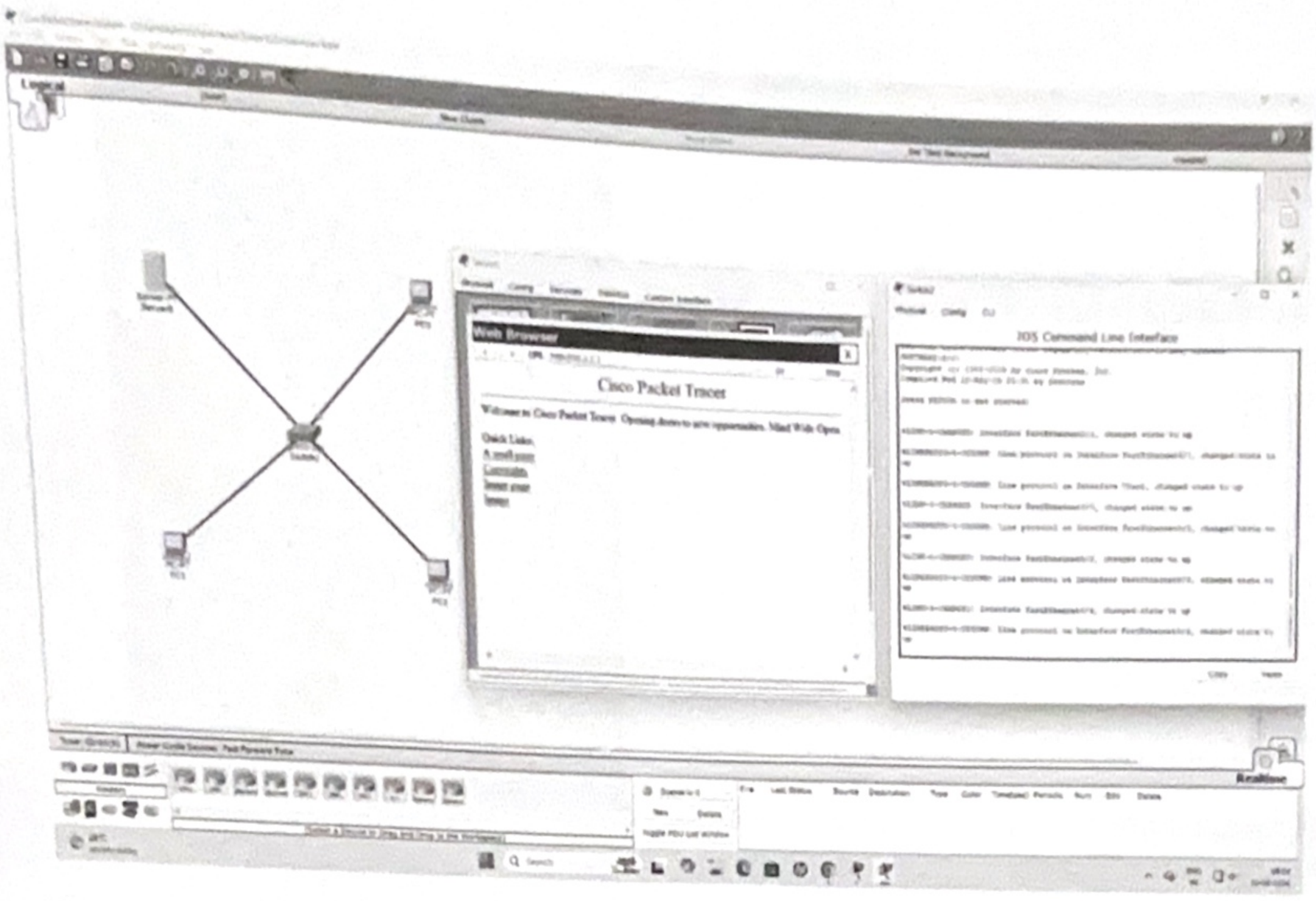
EXPERIMENT - 3





Experiment 4

Setup LAN



Result:

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP addresses within the same subnet (10.0.0.0/255.0.0.0). Ping tests between all devices returned successful replies, confirming end-to-end connectivity. The shared folder/server resource was successfully accessed from other computers on the network, confirming that the LAN was functioning correctly for resource sharing.

Experiment 6

Hamming Code

```python
def calc_r(m):
    return next(i for i in range(m) if 2**i >= m + i + 1)

def pos_r(data, r):
    j, k, res = 0, 1, ""
    for i in range(1, len(data) + r + 1):
        if i == 2**j:
            res += '0'
            j += 1
        else:
            res += data[-k]
            k += 1
    return res[::-1]

def calc_p(arr, r):
    n = len(arr)
    for i in range(r):
        val = sum(int(arr[-j]) for j in range(1, n + 1) if j & (2**i)) % 2
        arr = arr[:n-(2**i)] + str(val) + arr[n-(2**i)+1:]
    return arr

def detect(arr, r):
    res = sum(((sum(int(arr[-j]) for j in range(1, len(arr) + 1) if j & (2**i)) % 2) << i) for
    return res

# Execution
data = input("Enter the data:")
r = calc_r(len(data))
arr = calc_p(pos_r(data, r), r)
print("Data transferred:", arr)

err_arr =  input("Enter the error data:")
print("Error Data:", err_arr)

pos = detect(err_arr, r)
print("No error" if pos == 0 else f"Error position: {len(err_arr) - pos + 1} from left")
```

```
Input
Enter the data:1011001
Data transferred: 10101001110
Enter the error data:11101001110
Error Data: 11101001110
Error position: 3 from left

...Program finished with exit code 0
Press ENTER to exit console.
```

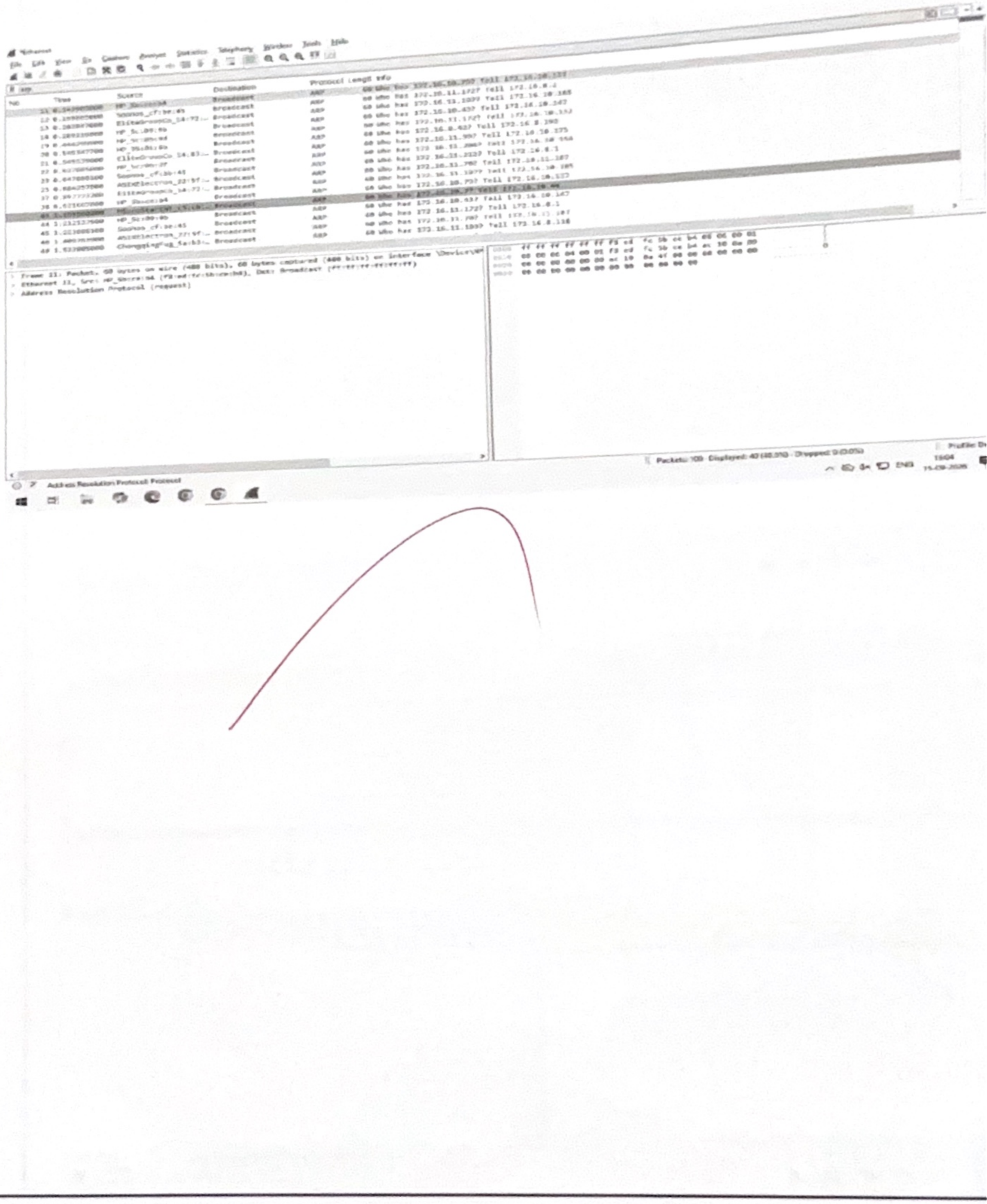
Result:

Given hamming code is successfully completed.

Experiment 5

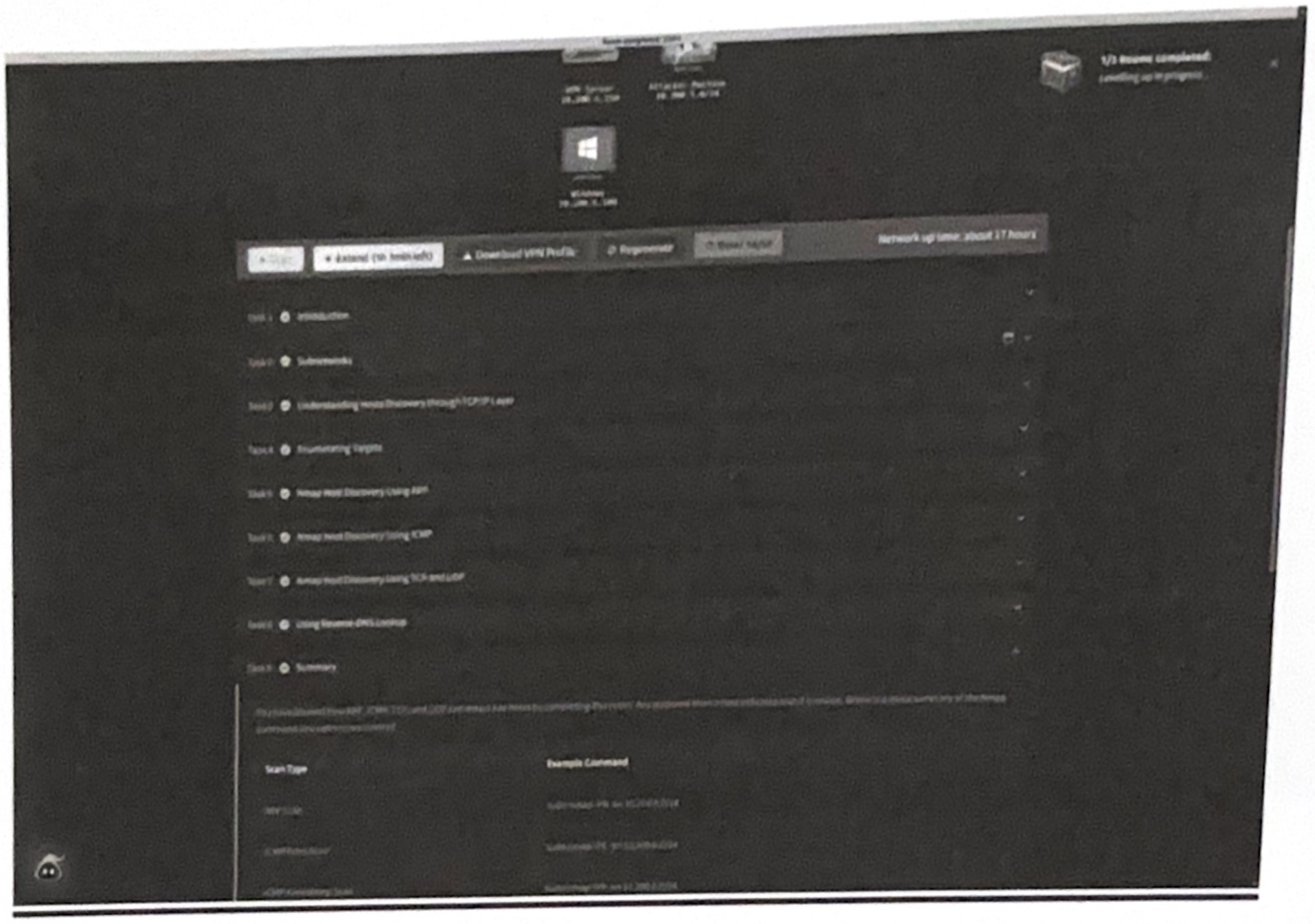
1. Create a Filter to display only TCP/UDP packets, inspect the packets and provide the flow graph

2. Create a Filter to display only ARP packets and inspect the packets.



Experiment 8

Nmap Using TryHackMe

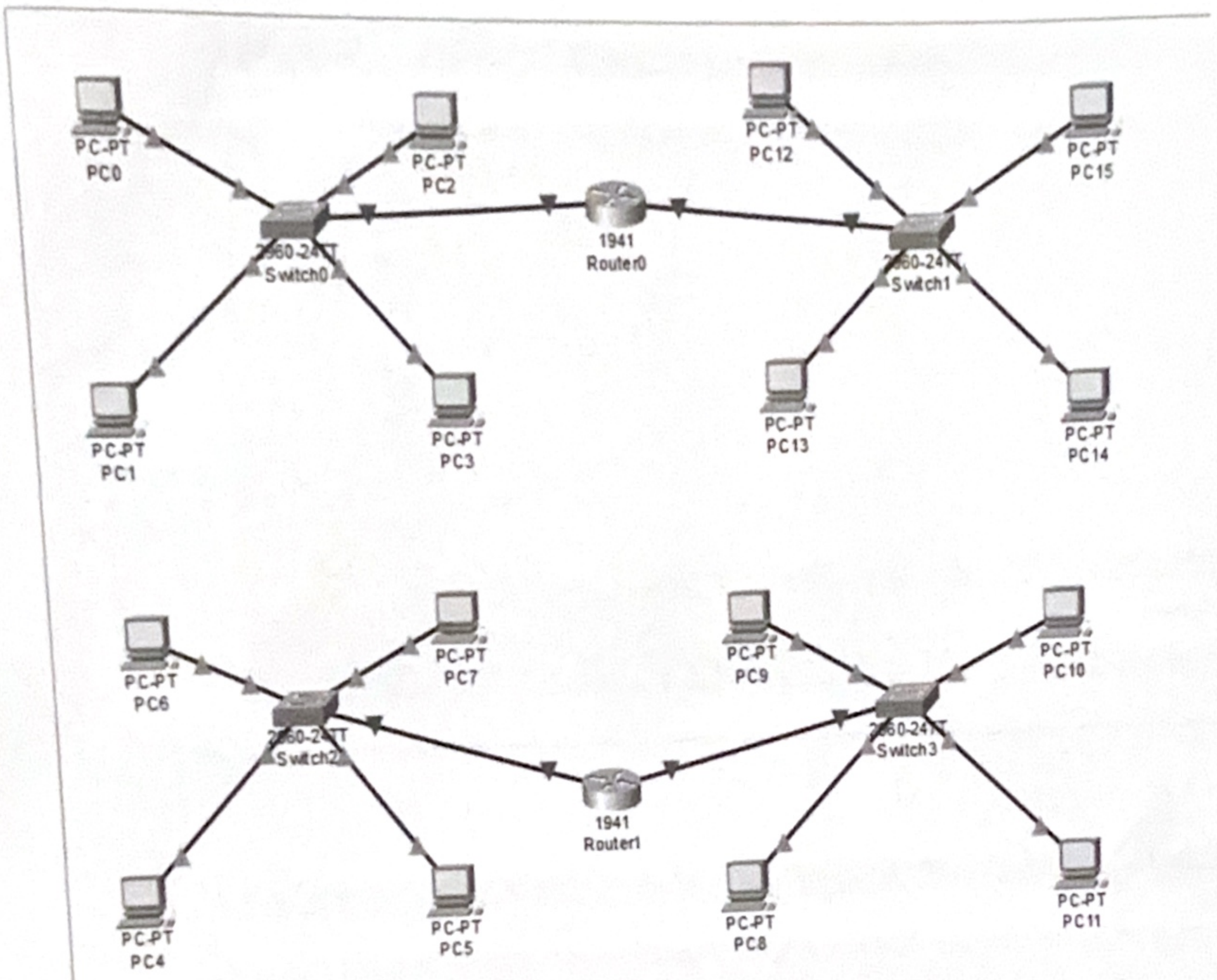


Result:

Given experiment is successfully completed.

Experiment 9

Subnetting



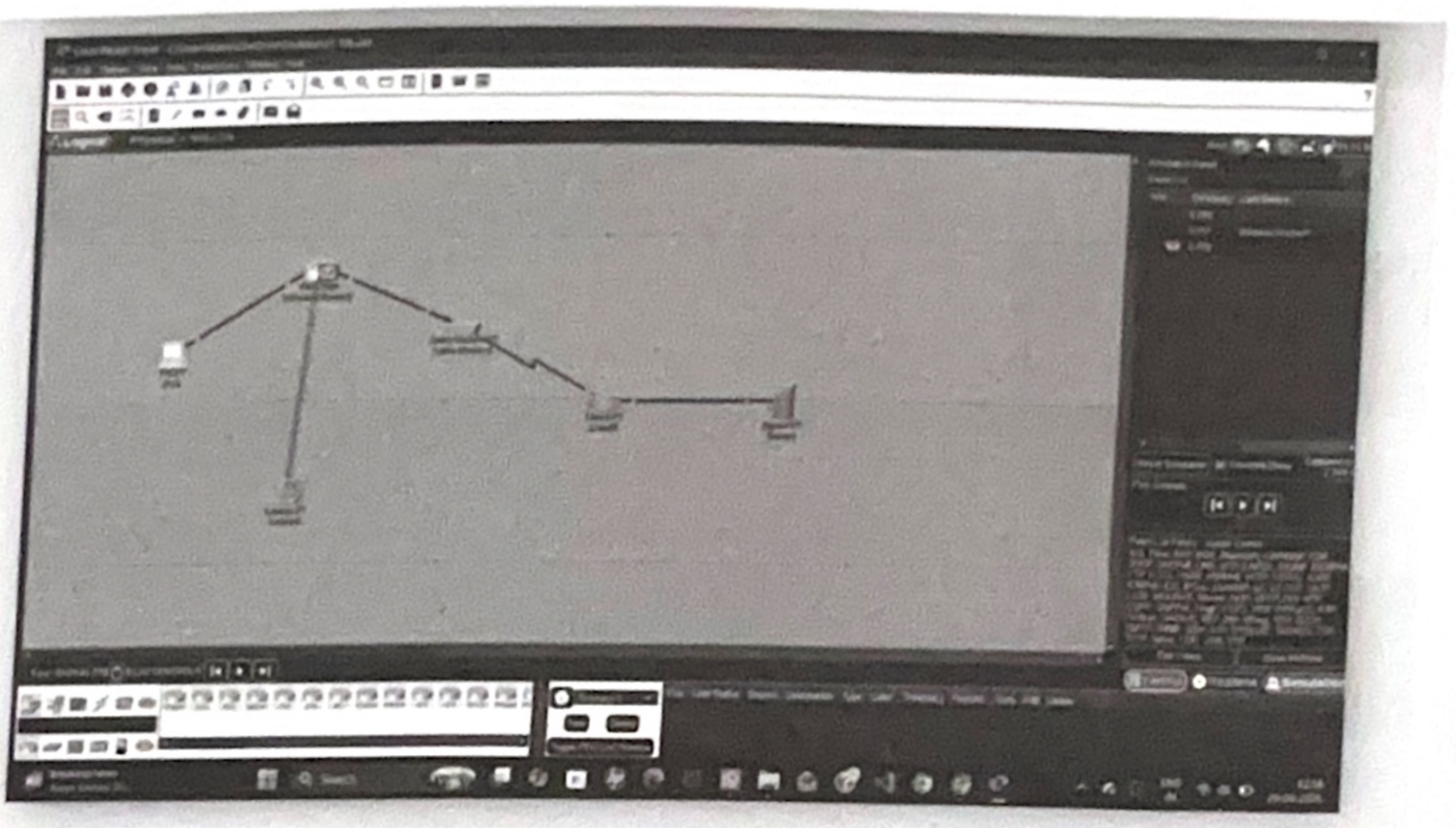
Result:

Given experiment is successfully completed.

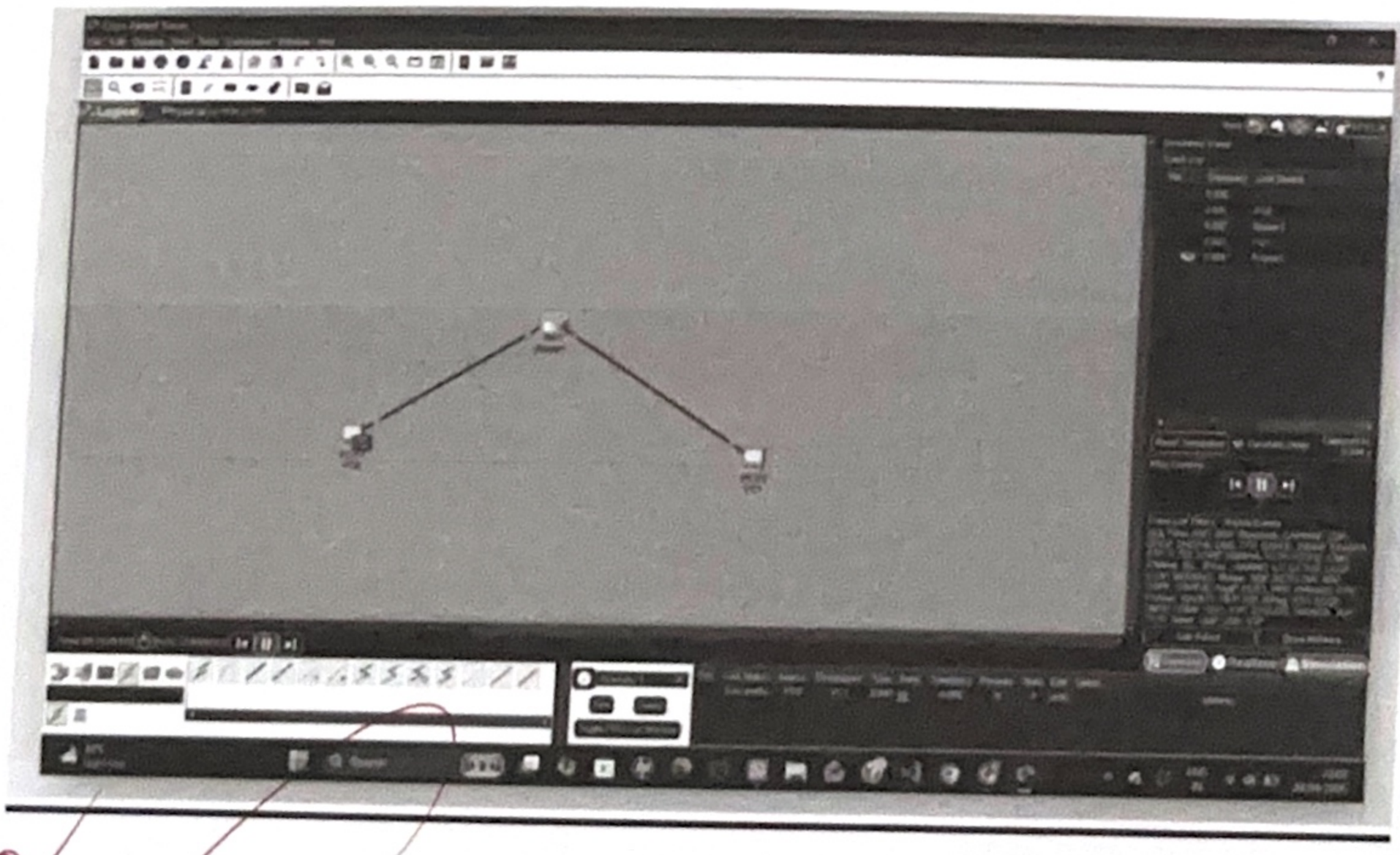
Experiment 10

Internetworking with Routers

a)



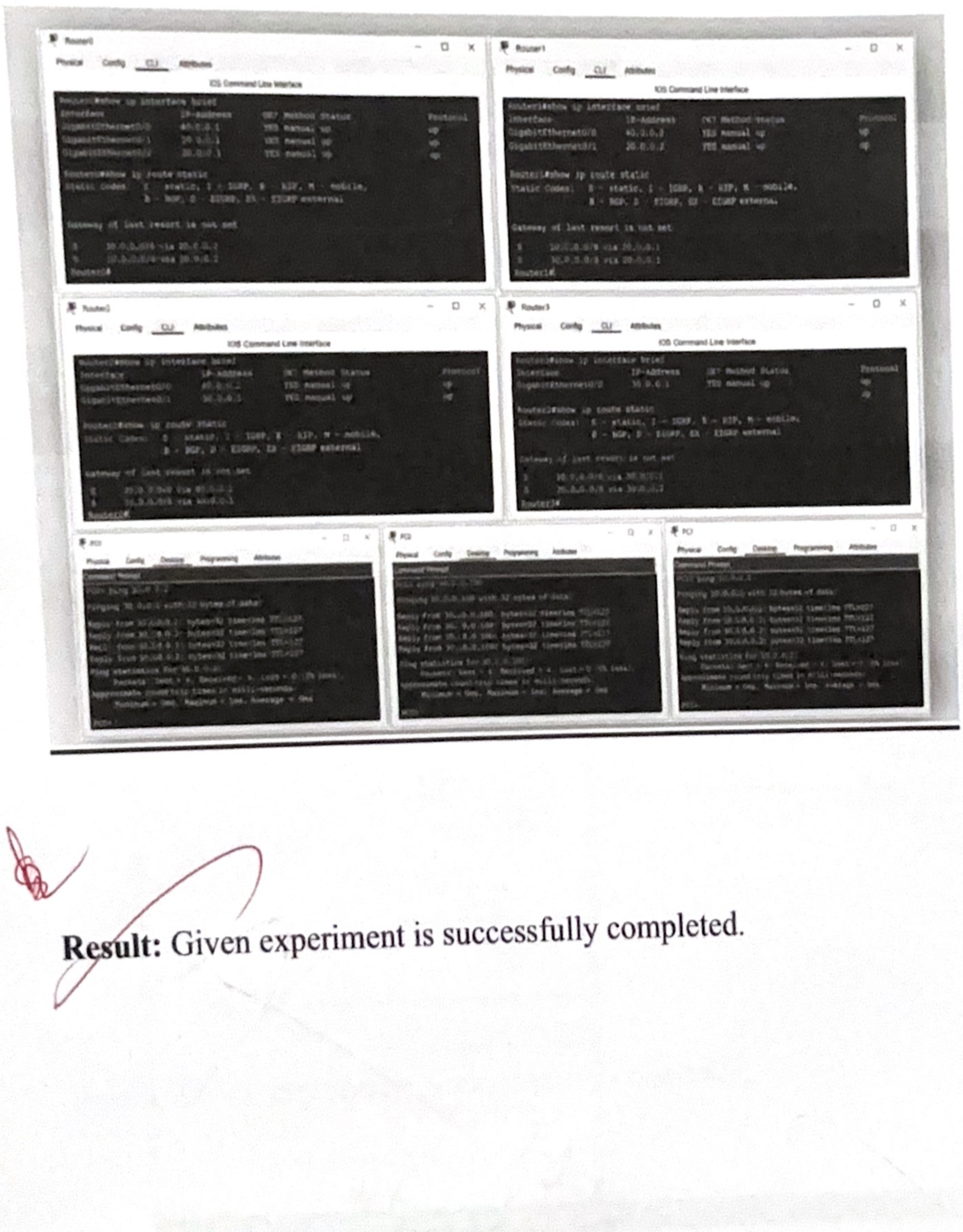
b)



Result: Given experiment is successfully completed.

Experiment 11

Static Routing Configuration



Result: Given experiment is successfully completed.

Experiment 12

Implement echo client server using TCP/UDP sockets.

```
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec  7 2023, 22:03:25) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\kishore\AppData\Local\Programs\Python\Python312\Ex 12(1) CN.py
Enter message: hello
Echo: hello
Enter message: exit
>>>
```

```
import socket
import threading
import time

def server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(("127.0.0.1", 5001))

    while True:
        data, addr = s.recvfrom(1024)
        if data.decode() == "exit":
            break
        s.sendto(data, addr)

    s.close()

threading.Thread(target=server, daemon=True).start()
time.sleep(1)

c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

while True:
    msg = input("Enter message: ")
    c.sendto(msg.encode(), ("127.0.0.1", 5001))
    if msg == "exit":
        break
    data, addr = c.recvfrom(1024)
    print("Echo:", data.decode())

c.close()
```

```
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec  7 2023, 22:03:25) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\kishore\AppData\Local\Programs\Python\Python312\Ex 12(1) CN.py
Enter message: hello
Echo: hello
Enter message: exit
>>>
```

```
import socket
import threading

def server():
    s = socket.socket()
    s.bind(("127.0.0.1", 5000))
    s.listen(1)
    c, addr = s.accept()

    while True:
        msg = c.recv(1024).decode()
        if msg == "exit":
            break
        c.send(msg.encode())

    c.close()
    s.close()

threading.Thread(target=server, daemon=True).start()

c = socket.socket()
c.connect(("127.0.0.1", 5000))

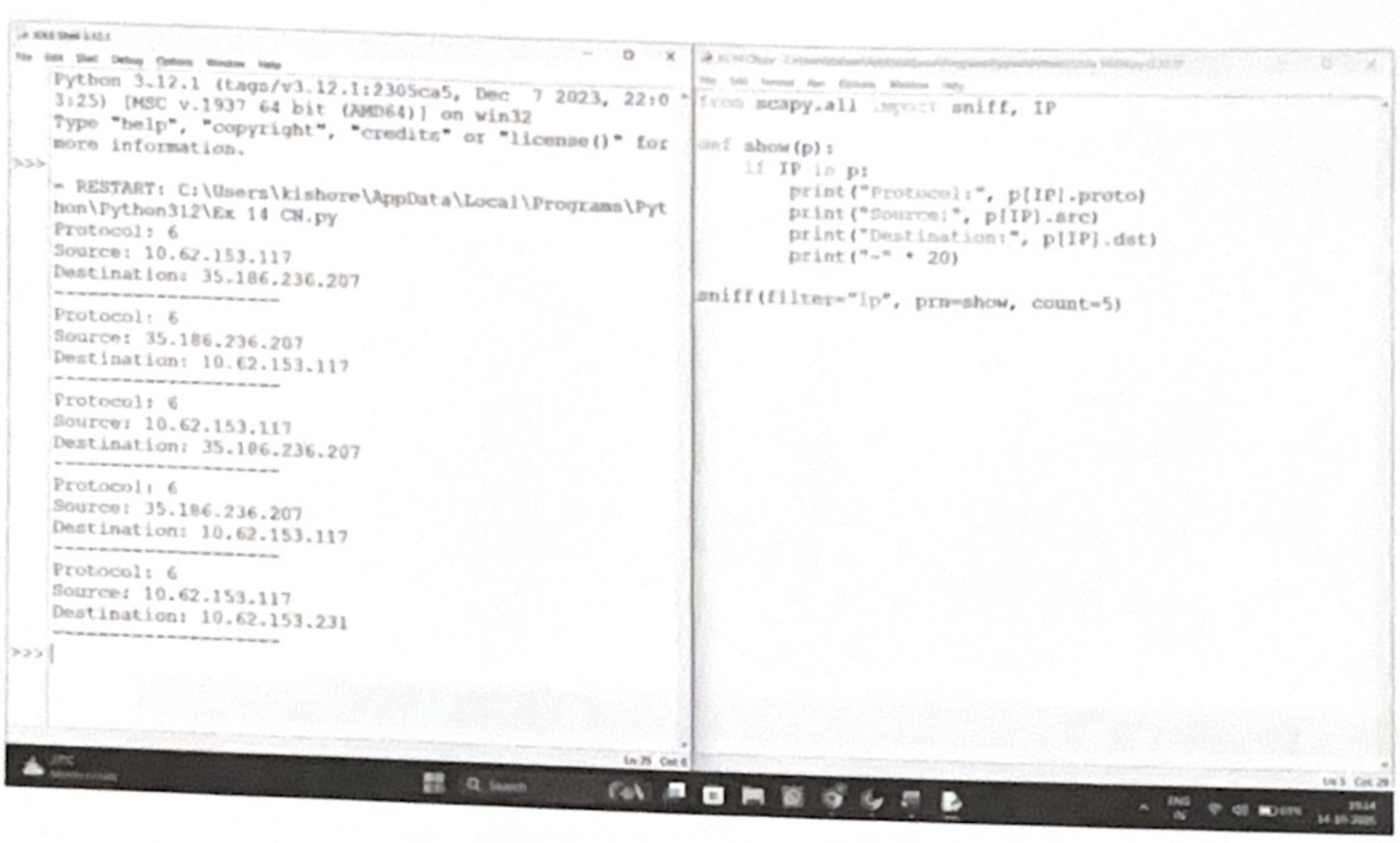
while True:
    msg = input("Enter message: ")
    c.send(msg.encode())
    if msg == "exit":
        break
    print("Echo:", c.recv(1024).decode())

c.close()
```

Result: Given experiment is successfully completed.

Experiment 14

Write a code using RAW sockets to implement packet sniffing.



Result: Given experiment is successfully completed.

Completed
15/10/24